msvtx_nTGCMiddleStation

overview · all regions and selections · rank 13 / 169

rank 13 / 169 by endcap / ht40 / D_equal

region	selection	D_equal	D_pooled	125/15	125/55	200/50	400/100	600/150	1000/275	n_bkg
endcap	all	0.601	0.683	0.504	0.471	0.511	0.672	0.754	0.755	34,970
endcap	ht40	0.589	0.673	0.483	0.445	0.494	0.662	0.743	0.746	7,390
endcap	bkg split	0.054	below 27,580 / above 7,390							
barrel	all	0.059	0.081	0.045	0.061	0.056	0.061	0.096	0.099	1,089
barrel	ht40	0.056	0.076	0.038	0.044	0.055	0.059	0.095	0.096	220
barrel	bkg split	0.042	below 869 / above 220							
all	all	0.381	0.435	0.336	0.318	0.334	0.449	0.466	0.471	36,059
all	ht40	0.371	0.427	0.326	0.325	0.324	0.439	0.458	0.463	7,610
all	bkg split	0.053	below 28,449 / above 7,610							

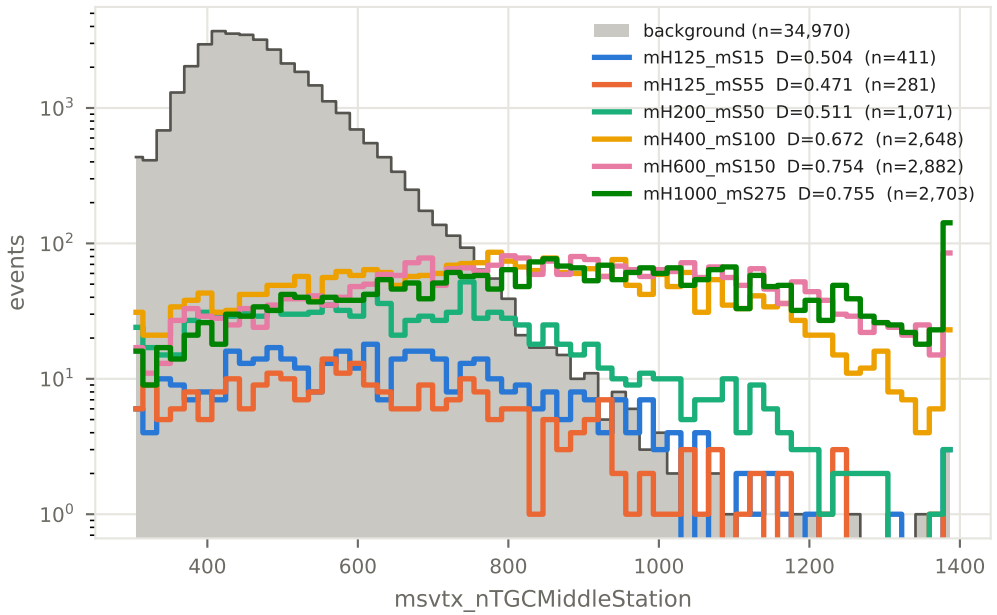
selections: all = every event; ht40 = htmissScalar_NOSYS >= 40 GeV, signal and background
pages: per region -> [all] 2x2, [ht40] 2x2, [background below vs above] 1x2

msvtx_nTGCMiddleStation

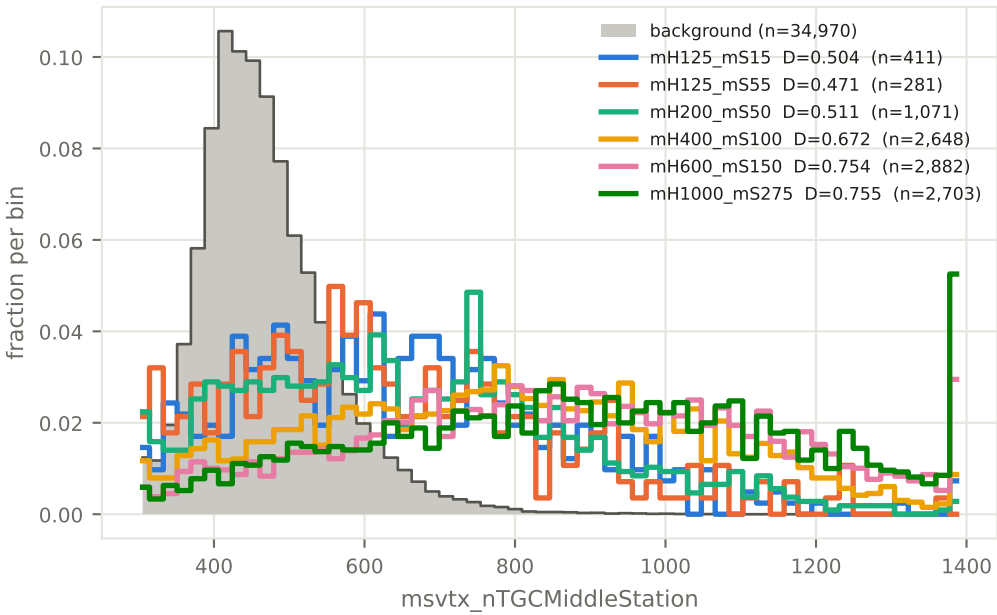
region: ENDCAP · selection: all (every event) · rank 13 / 169

D_equal 0.601 D_pooled 0.683 min 0.471 (mH125_mS55) max 0.755 (mH1000_mS275) bkg n = 34,970 sig n = 9,996

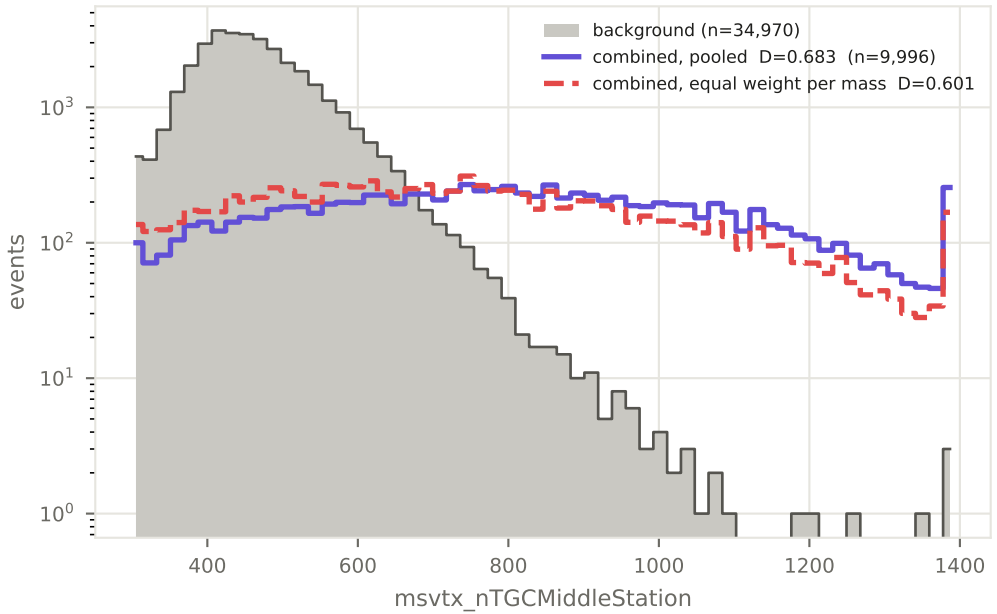
each mass point vs background · raw



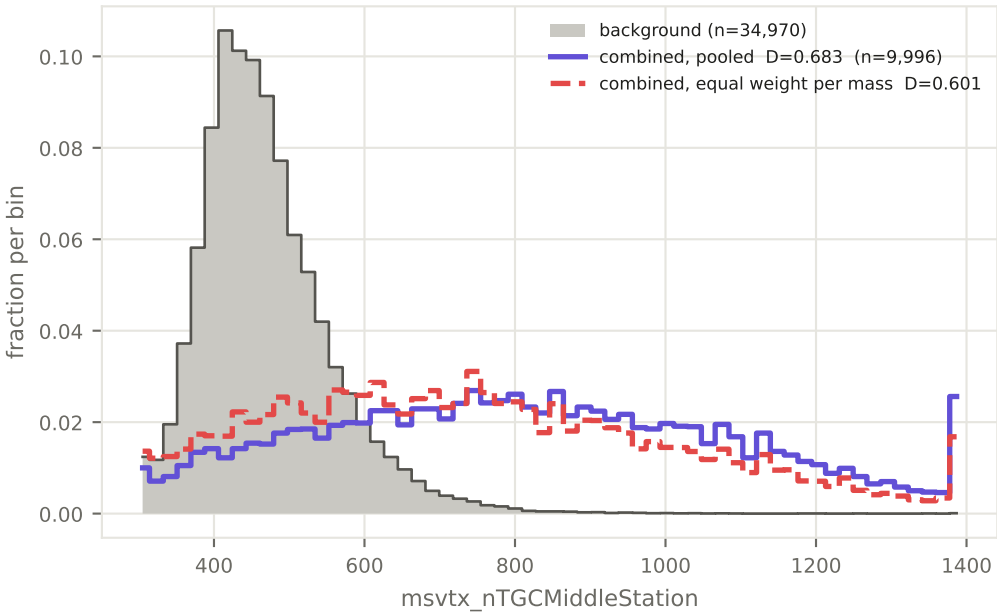
each mass point vs background · normalized



combined signal vs background · raw



combined signal vs background · normalized



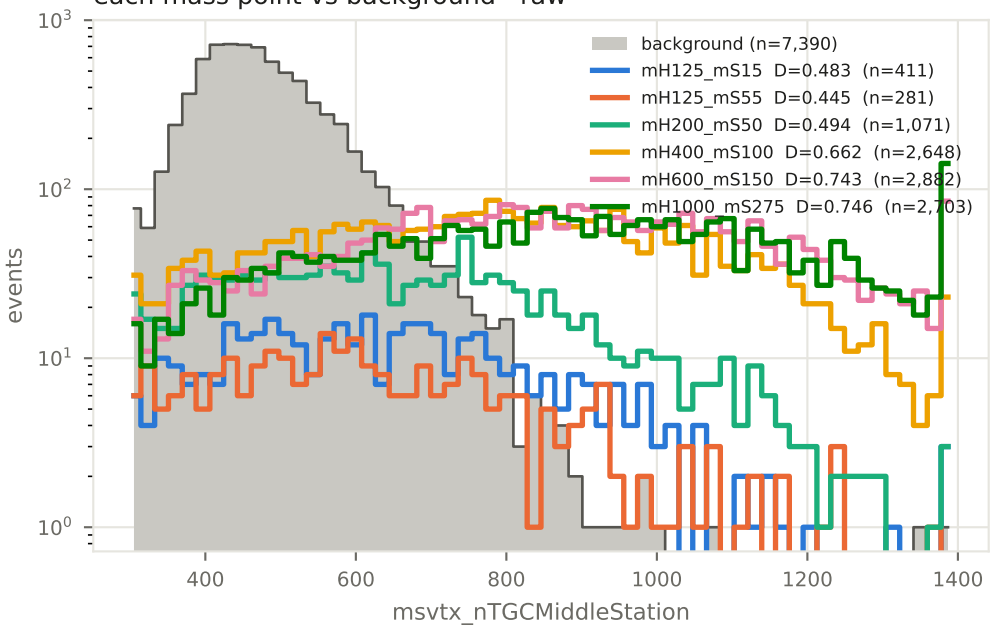
edge bins include under/overflow · NaN and -999 removed

msvtx_nTGCMiddleStation

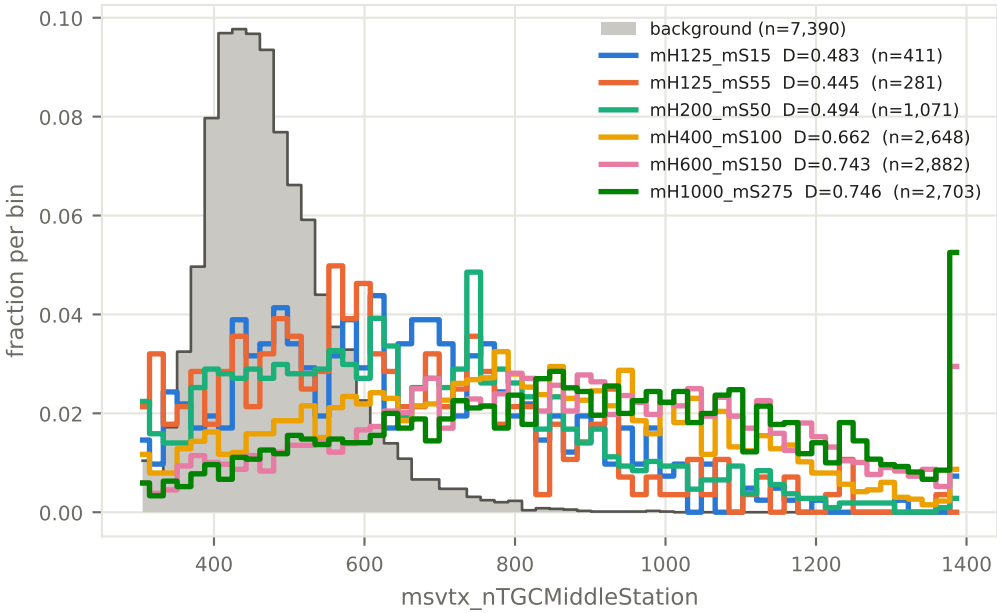
region: ENDCAP · selection: ht40 (htmissScalar_NOSYS >= 40 GeV, signal and background) · rank 13 / 169

D_equal 0.589 D_pooled 0.673 min 0.445 (mH125_mS55) max 0.746 (mH1000_mS275) bkg n = 7,390 sig n = 9,996

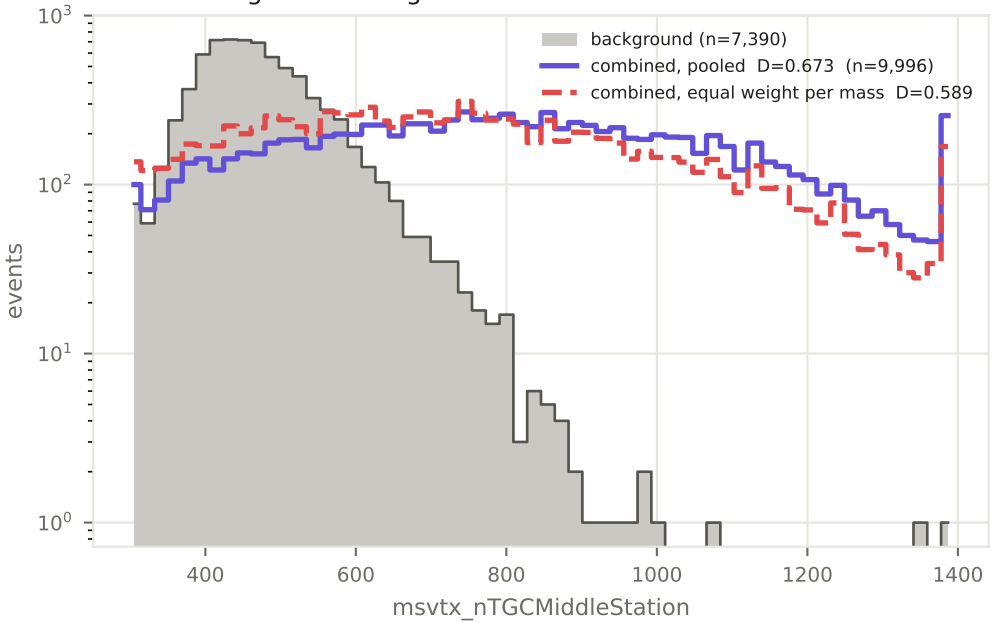
each mass point vs background · raw



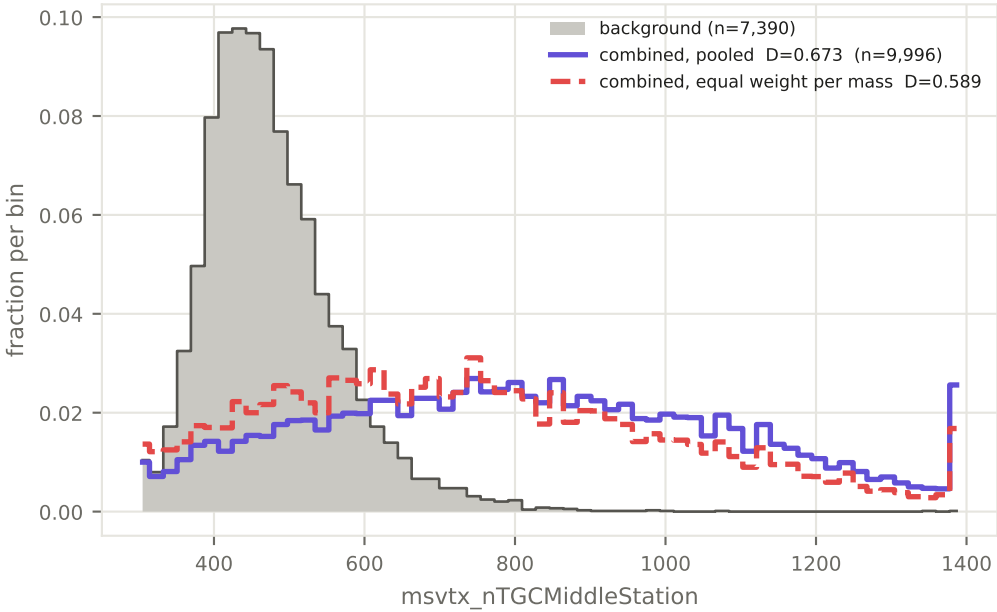
each mass point vs background · normalized



combined signal vs background · raw



combined signal vs background · normalized

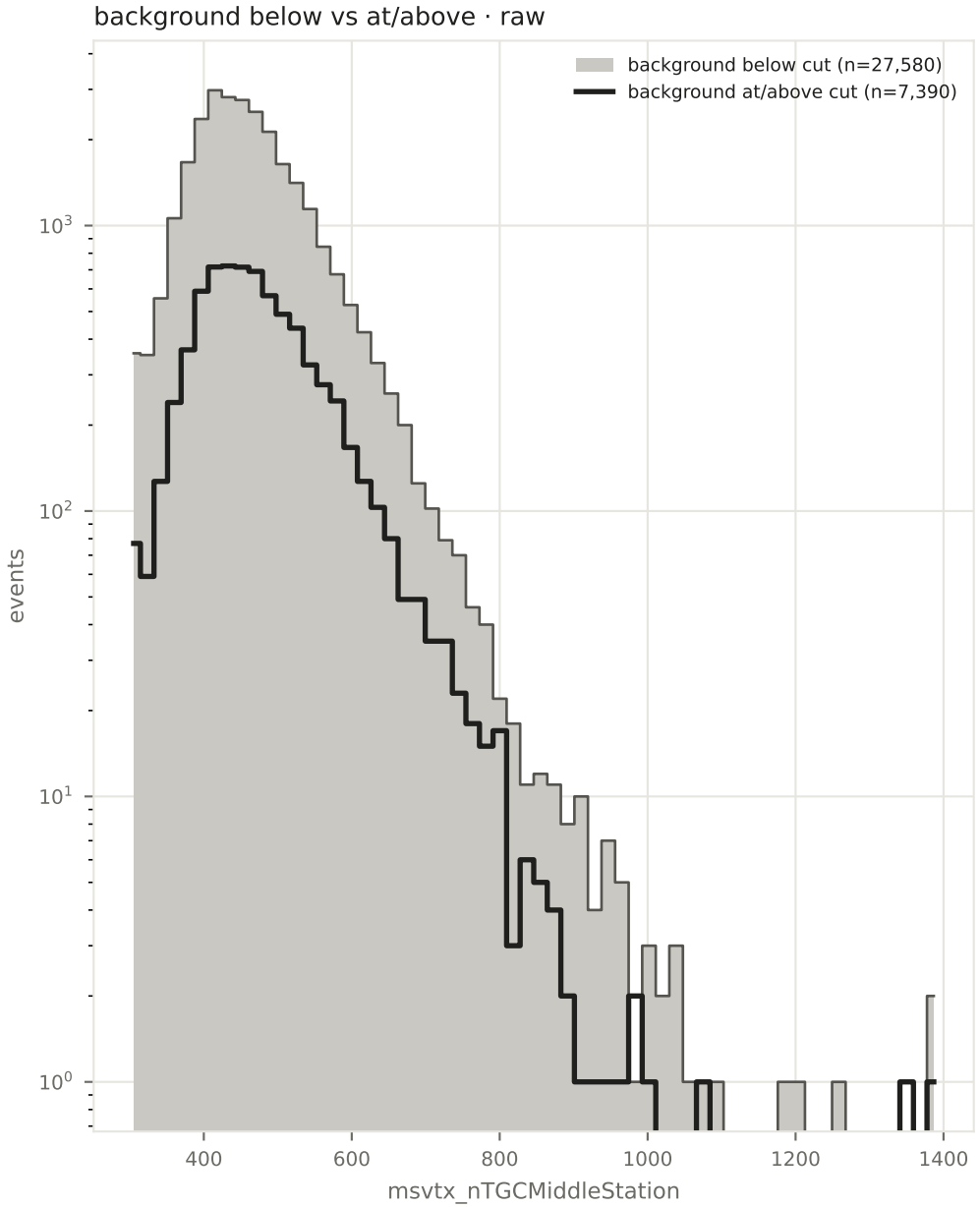


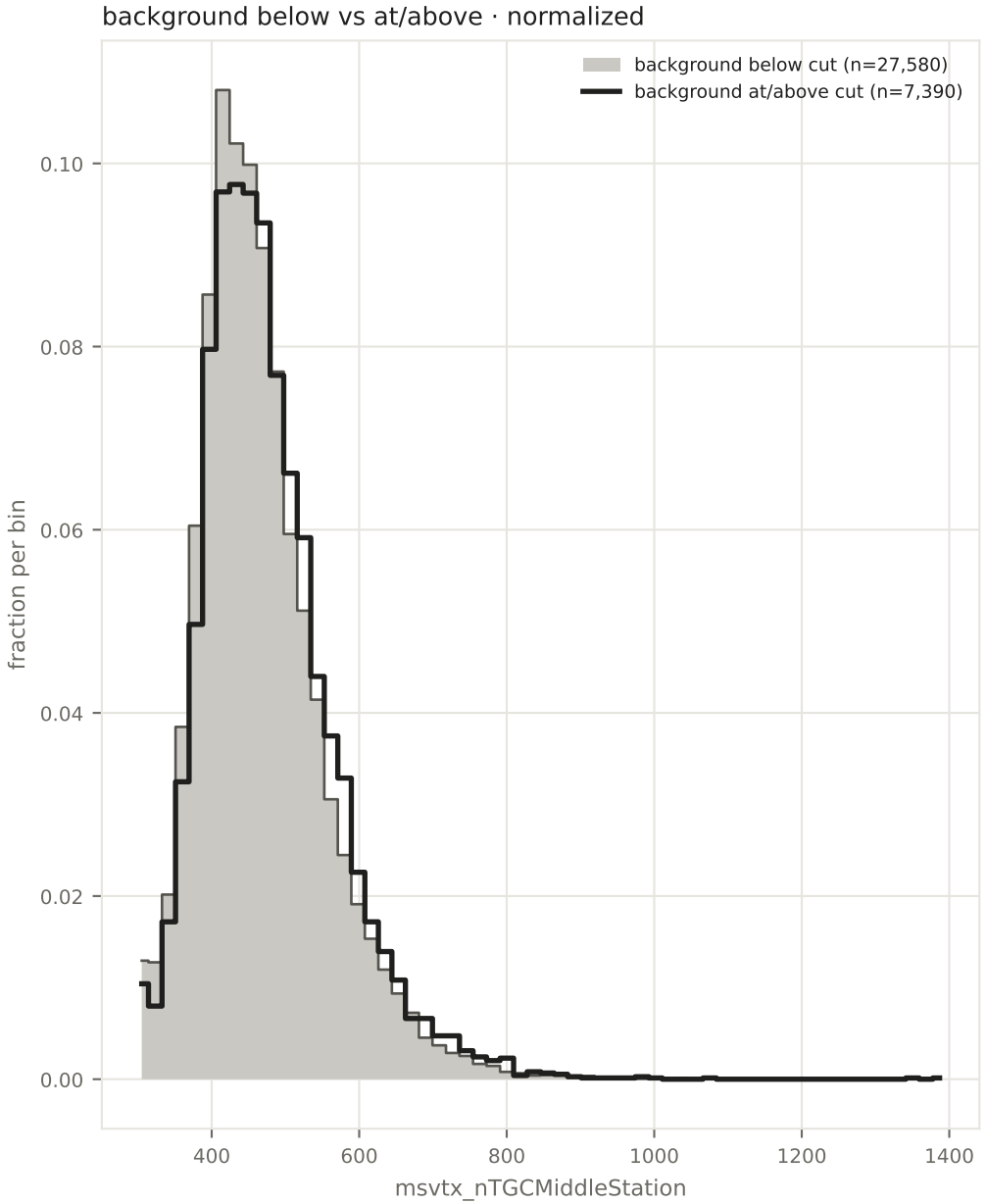
edge bins include under/overflow · NaN and -999 removed

msvtx_nTGCMiddleStation

region: ENDCAP · background only: below vs at/above the cut (htmissScalar_NOSYS >= 40 GeV, signal and background) · rank 13 / 169

D(below vs above) 0.054 n below = 27,580 n above = 7,390 large D = background shape changes across the cut



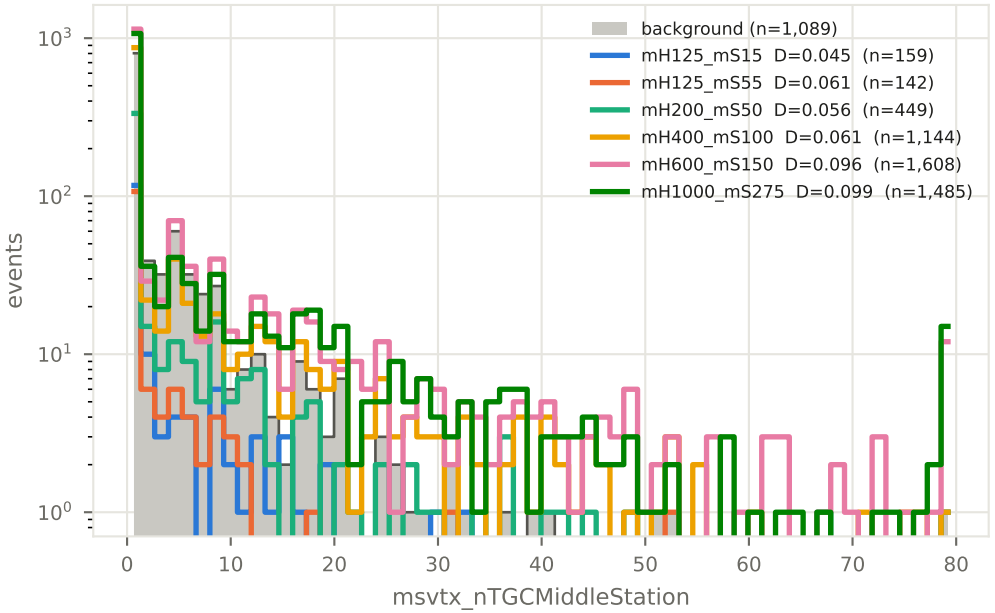


msvtx_nTGCMiddleStation

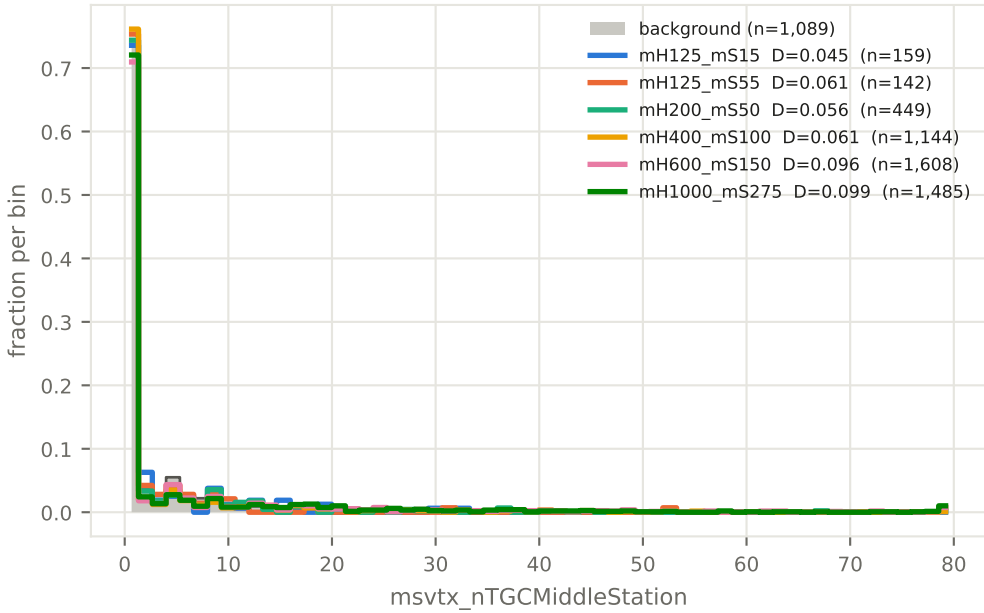
region: BARREL · selection: all (every event) · rank 13 / 169

D_equal 0.059 D_pooled 0.081 min 0.045 (mH125_mS15) max 0.099 (mH1000_mS275) bkg n = 1,089 sig n = 4,987

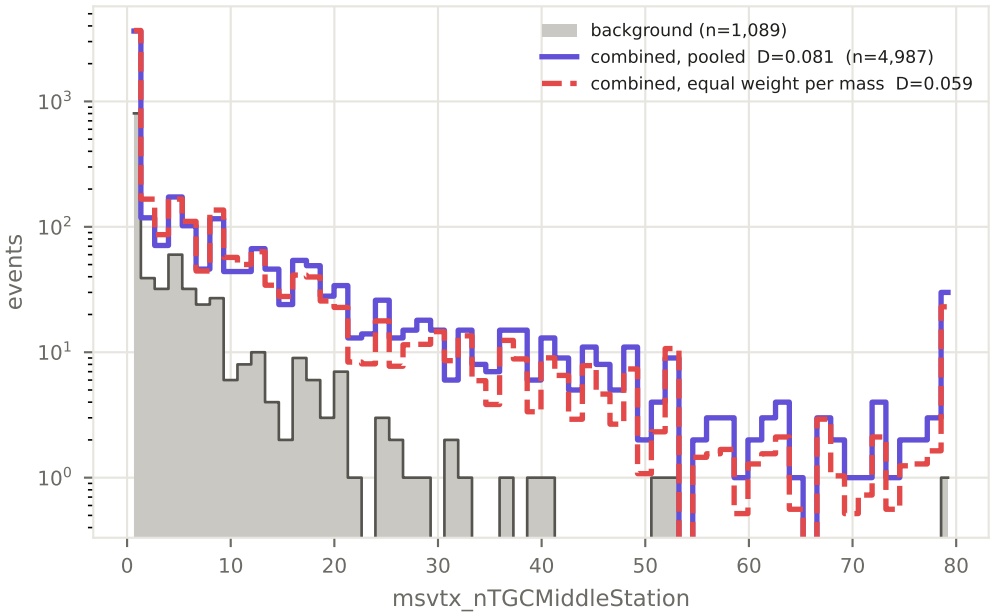
each mass point vs background · raw



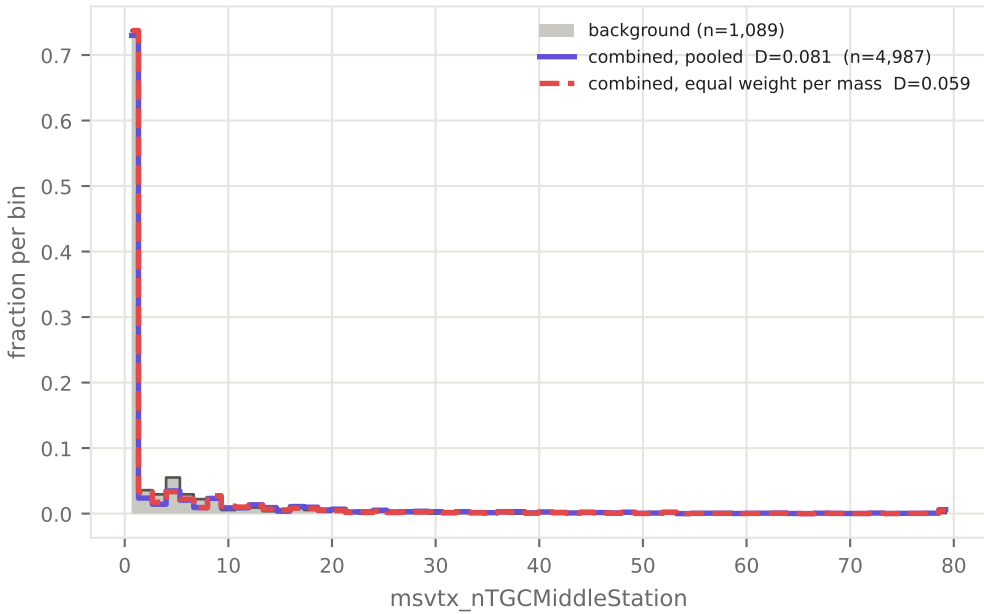
each mass point vs background · normalized



combined signal vs background · raw



combined signal vs background · normalized



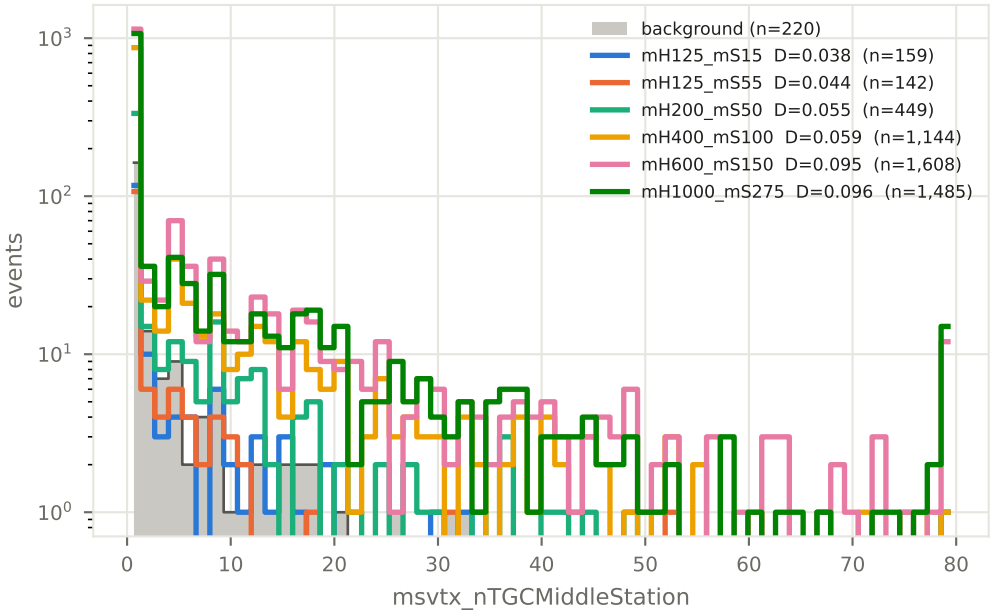
edge bins include under/overflow · NaN and -999 removed

msvtx_nTGCMiddleStation

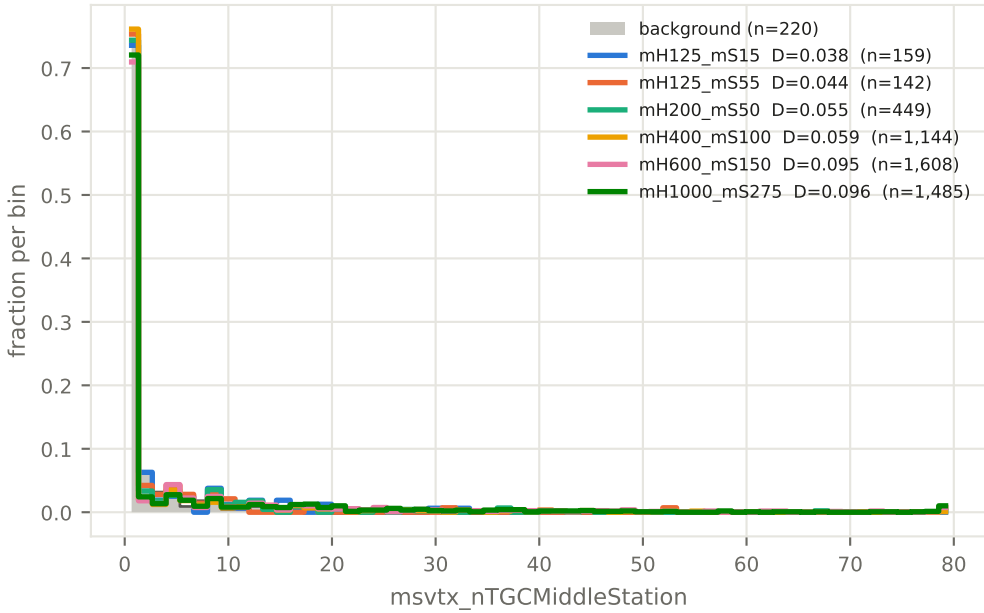
region: BARREL · selection: ht40 (htmissScalar_NOSYS >= 40 GeV, signal and background) · rank 13 / 169

D_equal 0.056 D_pooled 0.076 min 0.038 (mH125_mS15) max 0.096 (mH1000_mS275) bkg n = 220 sig n = 4,987

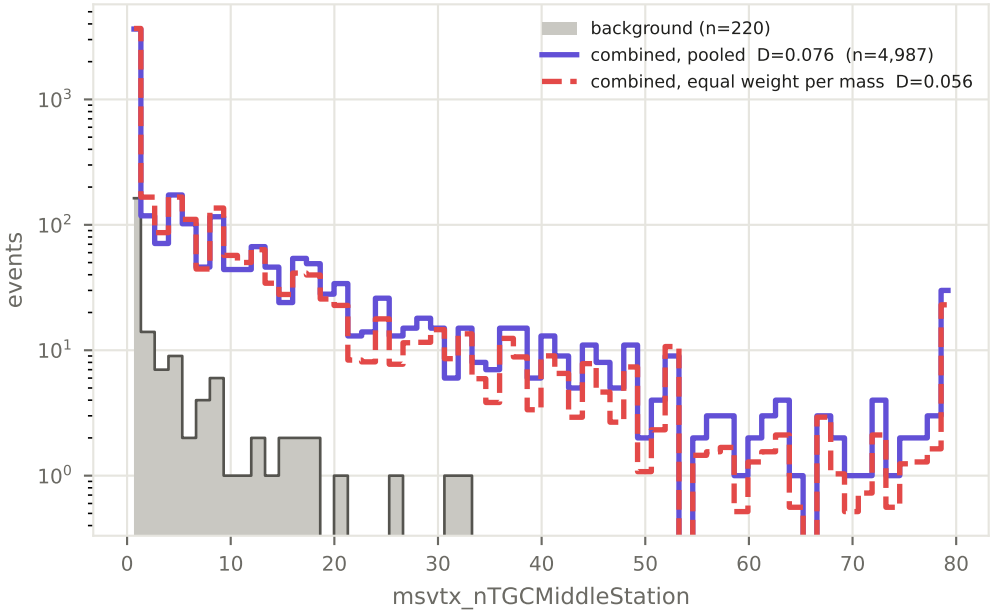
each mass point vs background · raw



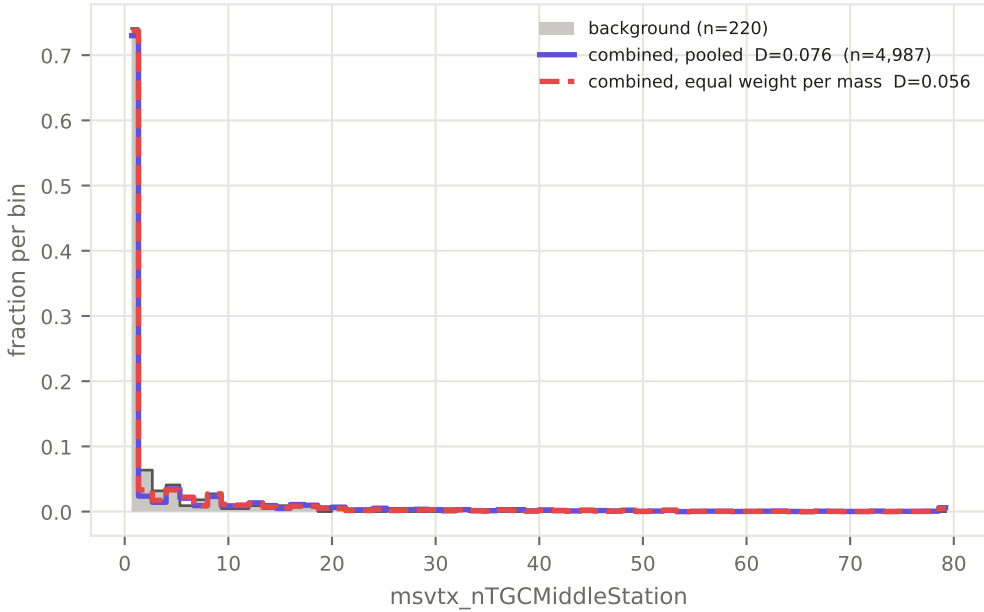
each mass point vs background · normalized



combined signal vs background · raw



combined signal vs background · normalized

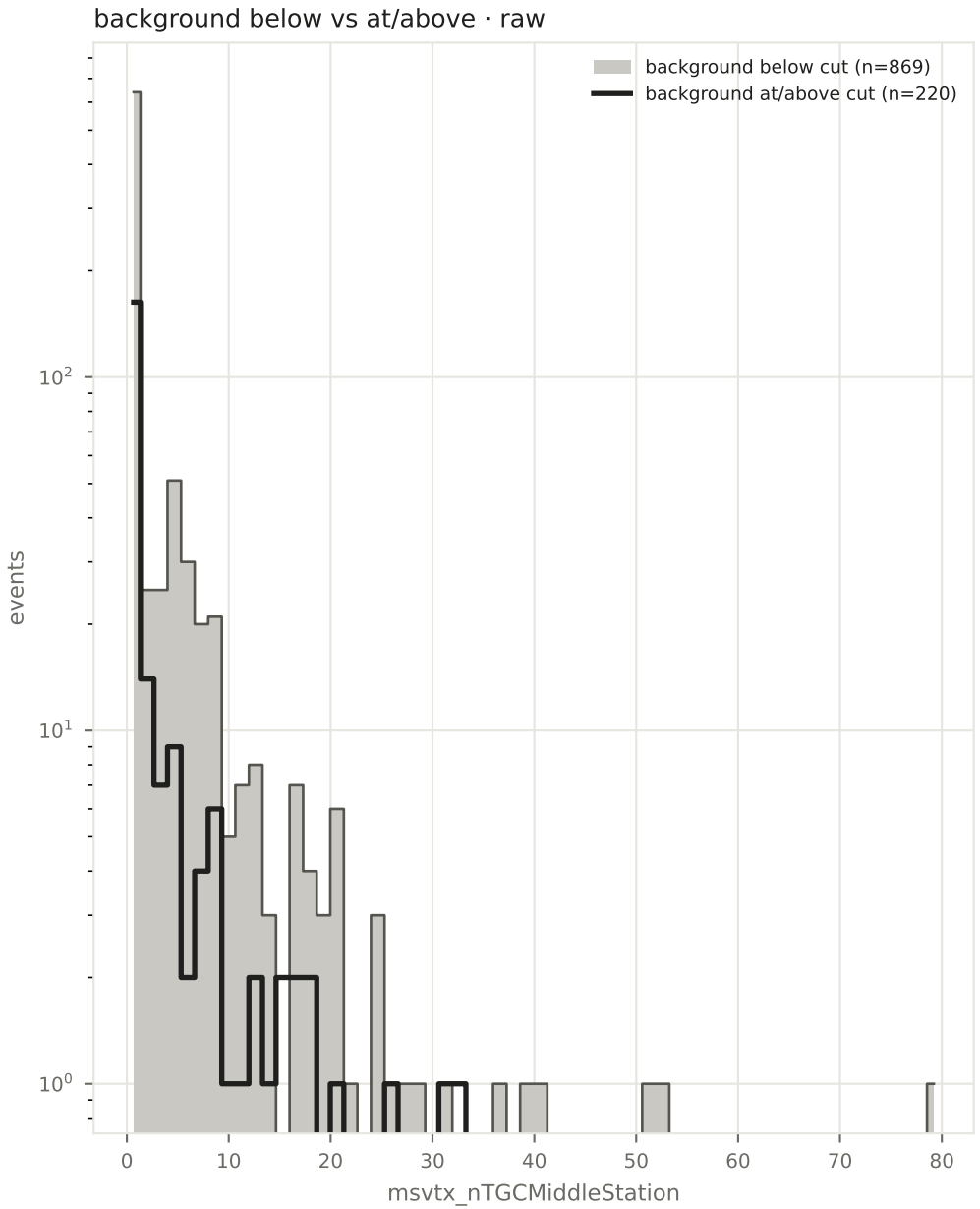


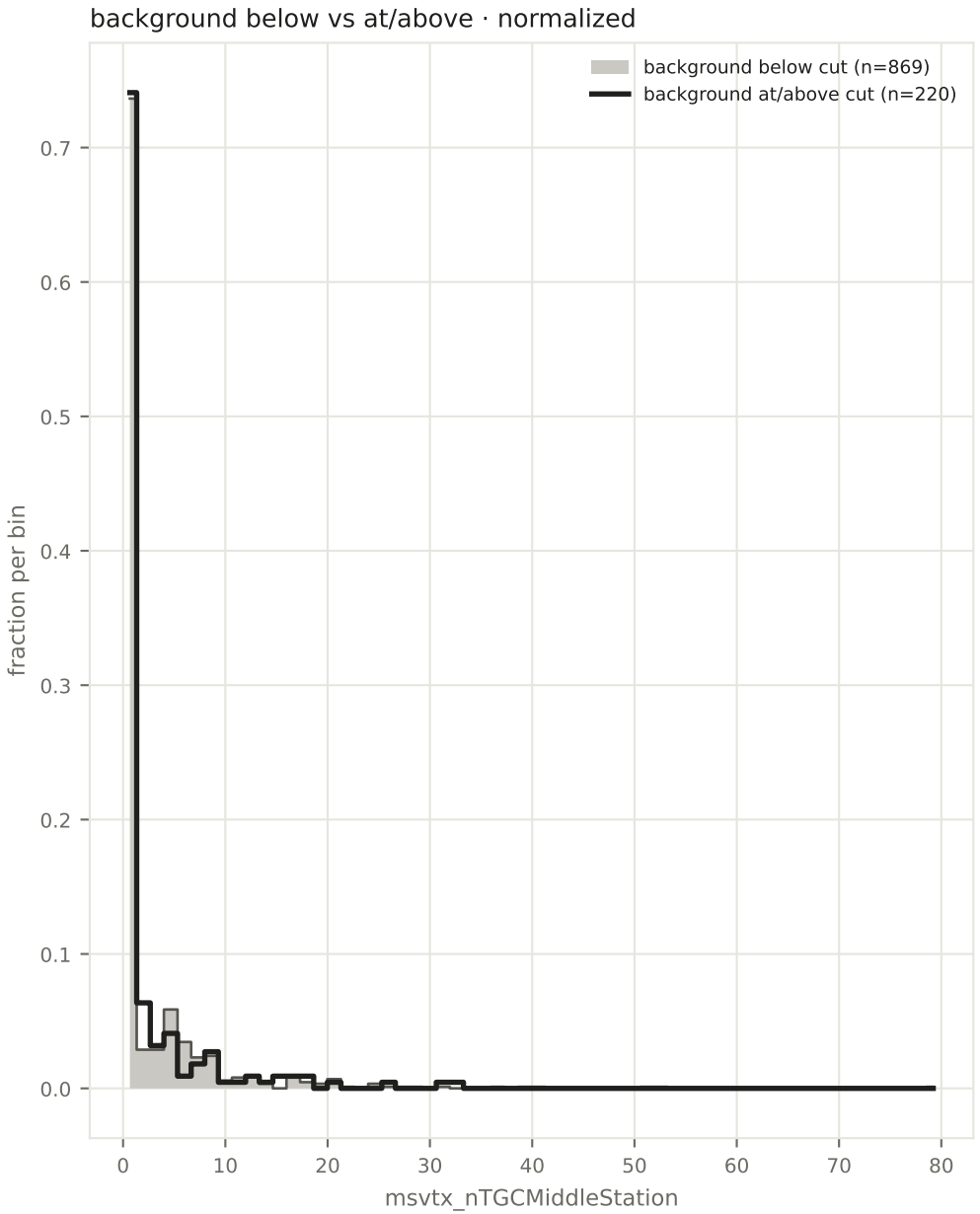
edge bins include under/overflow · NaN and -999 removed

msvtx_nTGCMiddleStation

region: BARREL · background only: below vs at/above the cut (htmissScalar_NOSYS >= 40 GeV, signal and background) · rank 13 / 169

D(below vs above) 0.042 n below = 869 n above = 220 large D = background shape changes across the cut



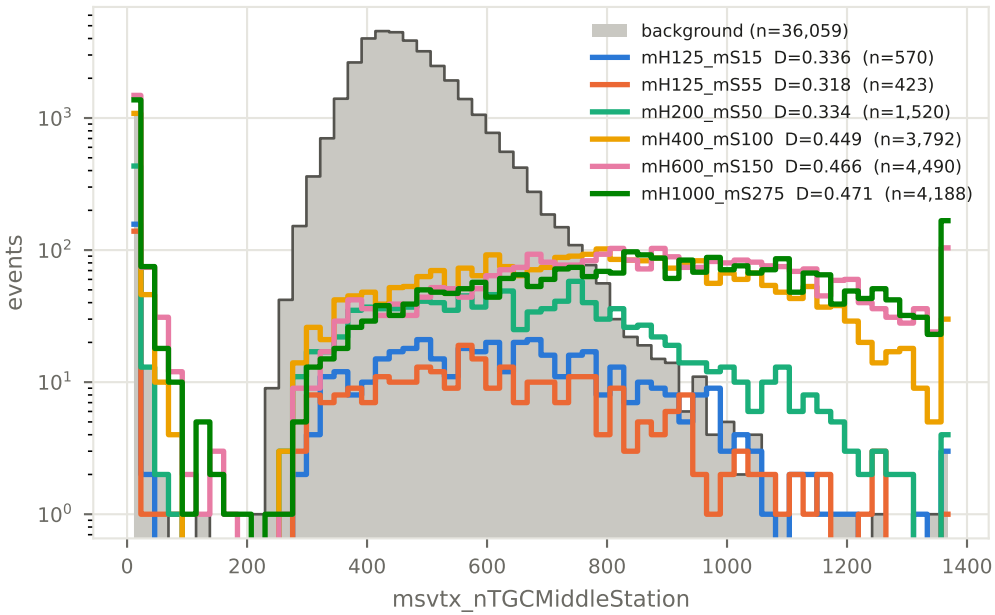


msvtx_nTGCMiddleStation

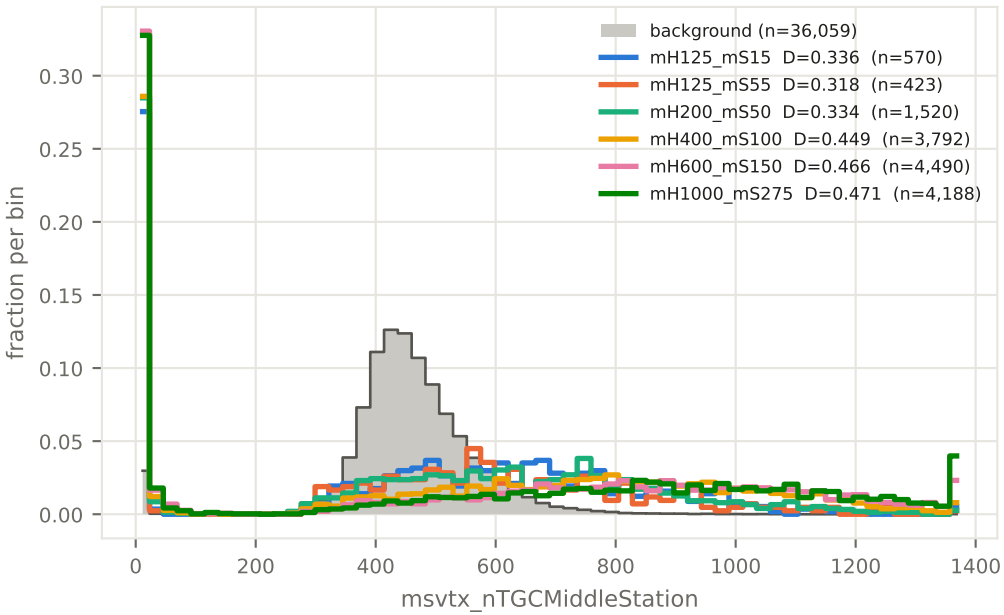
region: ALL · selection: all (every event) · rank 13 / 169

D_equal 0.381 D_pooled 0.435 min 0.318 (mH125_mS55) max 0.471 (mH1000_mS275) bkg n = 36,059 sig n = 14,983

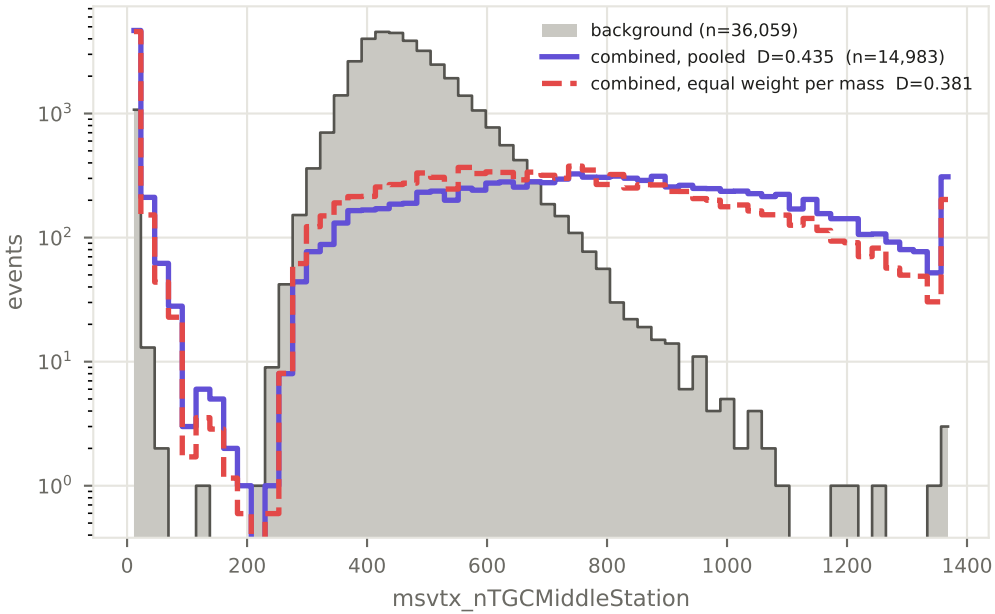
each mass point vs background · raw



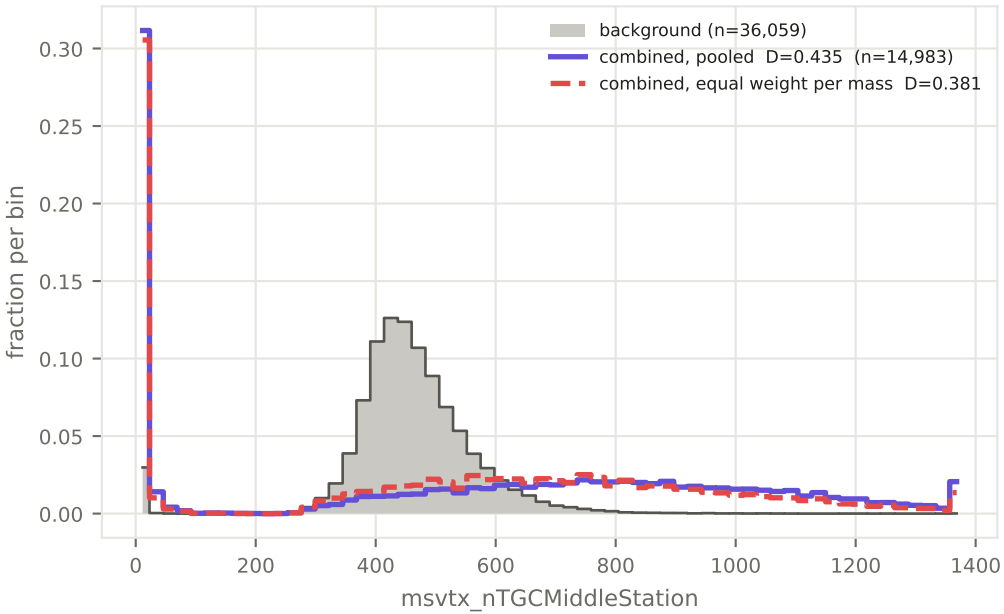
each mass point vs background · normalized



combined signal vs background · raw



combined signal vs background · normalized



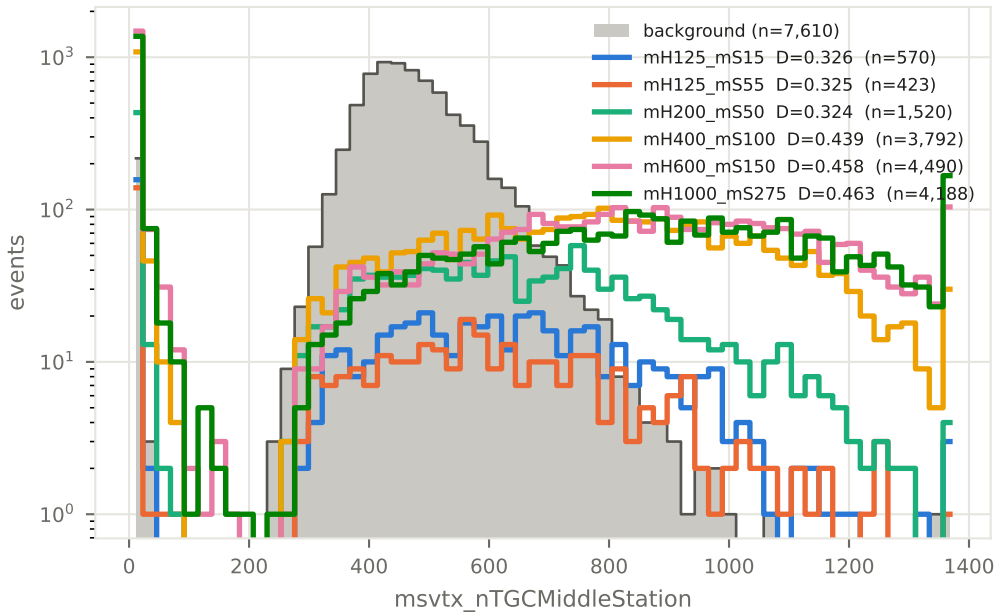
edge bins include under/overflow · NaN and -999 removed

msvtx_nTGCMiddleStation

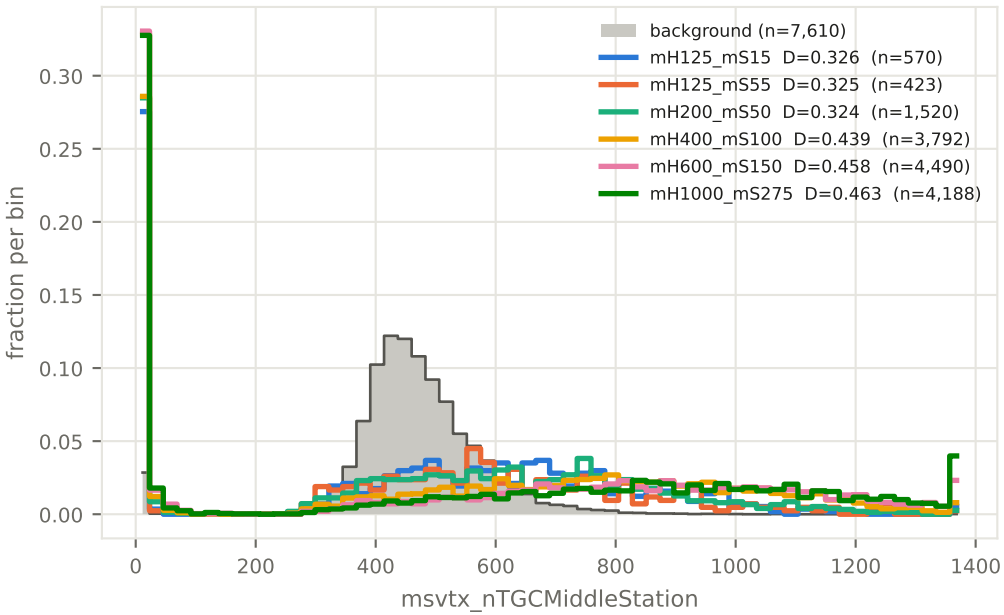
region: ALL · selection: ht40 (htmissScalar_NOSYS >= 40 GeV, signal and background) · rank 13 / 169

D_equal 0.371 D_pooled 0.427 min 0.324 (mH200_mS50) max 0.463 (mH1000_mS275) bkg n = 7,610 sig n = 14,983

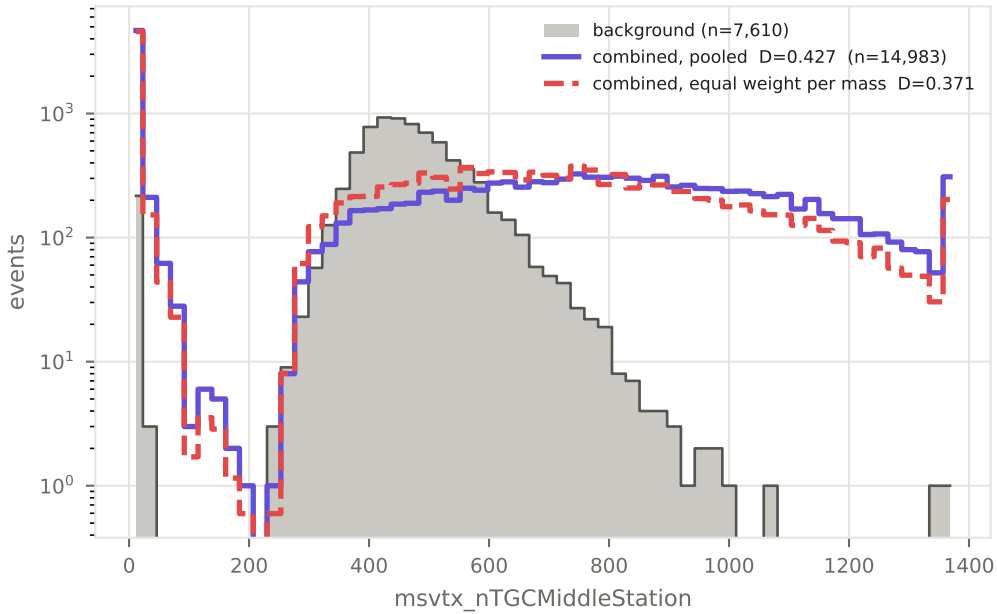
each mass point vs background · raw



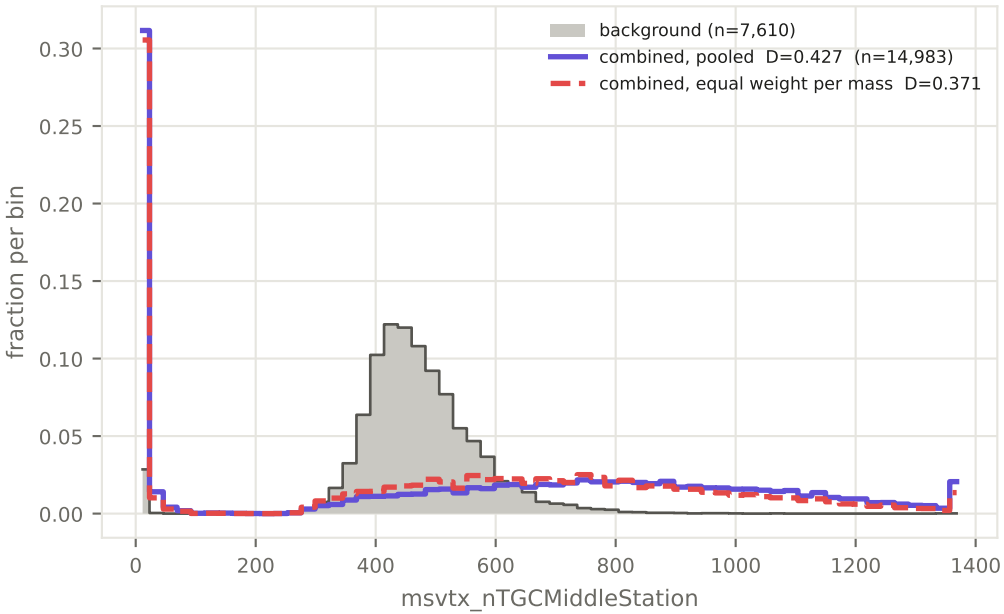
each mass point vs background · normalized



combined signal vs background · raw



combined signal vs background · normalized



edge bins include under/overflow · NaN and -999 removed

msvtx_nTGCMiddleStation

region: ALL · background only: below vs at/above the cut (htmissScalar_NOSYS >= 40 GeV, signal and background) · rank 13 / 169

D(below vs above) 0.053 n below = 28,449 n above = 7,610 large D = background shape changes across the cut

